

Checkpoint 7

Data 6560 Sport Analytics
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Risk vs. Reward: Analyzing Aggressive vs. Conservative Decision-Making in Golf

Quick Reminder

This project studies how aggressive play in golf relates to performance, using strokes gained metrics as a proxy. The central question is *“Do more ‘aggressive’ ball strikers consistently gain more strokes over the field, and how much of that edge comes from off the tee vs approach play?”*

The outcome modelled is SG_Total (total strokes gained per player-tournament). This matters for golfers and coaches deciding whether to pursue a more aggressive style or a more conservative approach.

Data

The same cleaned structured data is used as in checkpoint 6. This is a mix between the Masters 2022 sample and all of 2022 PGA tournament data.

Variables used are sg_total, sg_t2g, sg_putt, sg_ott, sg_app, bs_aggr, made_cut, score_par_diff

Filters put in place for row missing SG metrics (78 observations in Masters only data).

There were no changes made to the time window for checkpoint 7, instead the focus was on refining the model specification with the same outcome and core variables.

Checkpoint 6 Baseline

In Checkpoint 6, I used a simple OLS regression explained more in depth within CP6. The Masters only results had an R^2 of ~ 0.841 , an MAE ~ 0.61 strokes per player-tournament, and coefficients of $+1.11$ for bs_aggr and $+0.88$ for sg_putt.

The full 2022 season with $n = 29,180$ had an $R^2 \sim 0.853$ and coefficients of $+1.00$ for bs_aggr and $+1.01$ for sg_putt.

Baseline takeaway:

Across both the masters and full 2022 season, higher bs_aggr and better putting strongly predict higher sg_total. In the baseline model, a 1 unit increase in bs_aggr is worth about 1 stroke gained per event, which is a meaningful edge.

Upgrades in Checkpoint 7

For checkpoint 7, I made one focused upgrade:

- Bs_aggr is broken into its components, sg_ott (strokes gained off the tee) and sg_app (strokes gained on approach), while keeping bs_aggr and sg_putt in the model

The motivation for this was from the CP5 EDA and CP6 correlations, where bs_agr is highly correlated with sg_ott and sg_app. It is essentially ott + app. Coaches and players may care about which part of ball striking matters more:

- Is it mostly driving (sg_ott)
- Is it mostly approach shots (sg_app)
- Or both?

This upgrade connects directly to the idea of aggressiveness since aggressive drivers off the tee vs aggressive iron play may have different payoffs.

So, the CP7 model tests whether adding sg_ott and sg_app will:

1. Improve predictive performance (R^2 , MAE)
2. Reveal more broken down/interpretable effects

Analysis Spec

	CP6	CP7
Element	Baseline Spec	Upgrade Spec
Outcome	SG_Total	SG_Total
Inputs	bs_aggr, sg_putt	bs_aggr, sg_putt, sg_ott, sg_app
Sample	2022 Masters	2022 Masters
Row	player x tournament	player x tournament
Expected signs	Both positive (+)	All four positive (+)

Results and Comparison

Masters 2022 baseline vs upgraded analysis

Model	Inputs	R^2	MAE (strokes)	bs_aggr coef	sg_ott coef	sg_app coef	sg_putt coef
CP6 Baseline	bs_aggr, sg_putt	0.841	0.608	1.11	—	—	0.88
CP7 Upgrade	bs_aggr, sg_putt, sg_ott, sg_app	0.842	0.608	0.75	0.4	0.35	0.88

The practical interpretation of this comparison is that predictive accuracy (R^2 and MAE) does not change much, which is expected since bs_aggr is built from OTT and APP

However, the interpretation improves:

- In CP6 bs_aggr alone had a 1.11 stroke effect per unit
 - In CP7 that effect is spread out, 0.75 remains with bs_aggr, 0.40 with sg_ott, 0.35 with sg_app
- This supports the idea that both driving and approach contribute meaningfully to aggressive success at the Masters

A stability check was also done on the 'made cut' subset. This was done by restricting the sample to only players who made the cut:

Model	Sample	R ²	MAE (strokes)	bs_aggr coef	sg_ott coef	sg_app coef	sg_putt coef
Baseline	Made cut only	0.799	0.507	1.16	—	—	0.99
Upgrade	Made cut only	0.808	0.495	0.81	0.54	0.27	0.97

The takeaway of this stability check is that among players who played all four rounds, the upgrade increases R squared from 0.799 to 0.808 and reduces the MAE from 0.51 to 0.49 strokes.

For core contenders, decomposing aggressiveness into OTT + APP yields modestly better predictions and similar coefficient patterns.

Checks

- No leakage (all inputs computed from same tournament, no external info)
- Scale check (all variables measured in strokes, coefficients directly interpretable in strokes gained)
- *Multicollinearity*:
 - The design shows strong collinearity due to $bs_aggr \sim sg_ott + sg_app$
 - This is noticed by a large condition number in the regression output
 - While this doesn't break the predictions, it means individual coefficient magnitudes should be cautiously interpreted
- Subgroup check:
 - Made_cut_only subset shows similar coefficient signs and slightly better fit, indicating the model behaves similarly for higher performing (cut-making) players

Interpretation

These results still show that aggressive ball strikers perform better. Even after adding strokes gained off the tee and strokes gained on approach, higher ball striking aggressiveness is associated with better strokes gained total. The coefficient shrinks due to part of its effect measured by OTT and APP separately

Both driving and approach shots matter (OTT and APP) since they each have positive, significant contributions (0.40 and 0.35 strokes per unit), which confirms both aggressive driving and aggressive approaches contribute to outperforming the field.

Putting remains a strong multiplier with a coefficient $\sim +0.9$. This means better putting consistently boosts outcomes, regardless of aggressiveness or ball striking style. This makes sense logically, especially at a pro level.

For players who made the cut, the upgraded model is slightly more accurate (MAE down 0.02 strokes per player-tournament), which reinforces the idea that aggression plus putting is a winning formula.

Since this is a lot of words, an even simpler way to look at it is by analyzing the top golfer in the world, Scottie Scheffler, who exhibits these characteristics. He is a slightly above average putter

compared to the field – however he is one of the best and most aggressive ball strikers off the tee and from the fairway. He gives himself easier putts by taking more aggressive (and riskier) shots on the rest of the course.

Limits

- There is a moderate redundancy in metrics since `bs_aggr` is constructed from `sg_ott` + `sg_app`. With all 3 in the same model there is multicollinearity. The results are more interpretable but not ideal for a final predictive model
- The CP7 upgrade is implemented on the 2022 Masters data only, giving it a singular tournament specific context

Next Steps

For checkpoint 8, I plan to:

- Simplify the final model specification
- Re-fit on the full 2022 season
- Translate the effects into coaching and practicing recommendations